MATH329 · OPERATIONS RESEARCH

Midterm 462 Solutions

09/03/2025 · 30 marks

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01 Solution Q1: theory

a The special cases of the graphical method

1. Alternative optimal solutions
2. Unbounded solution
3. No feasible solution

b The two requirements to start simplex

1. The coefficients of the basic variables (slack variables) form an Identity matrix.

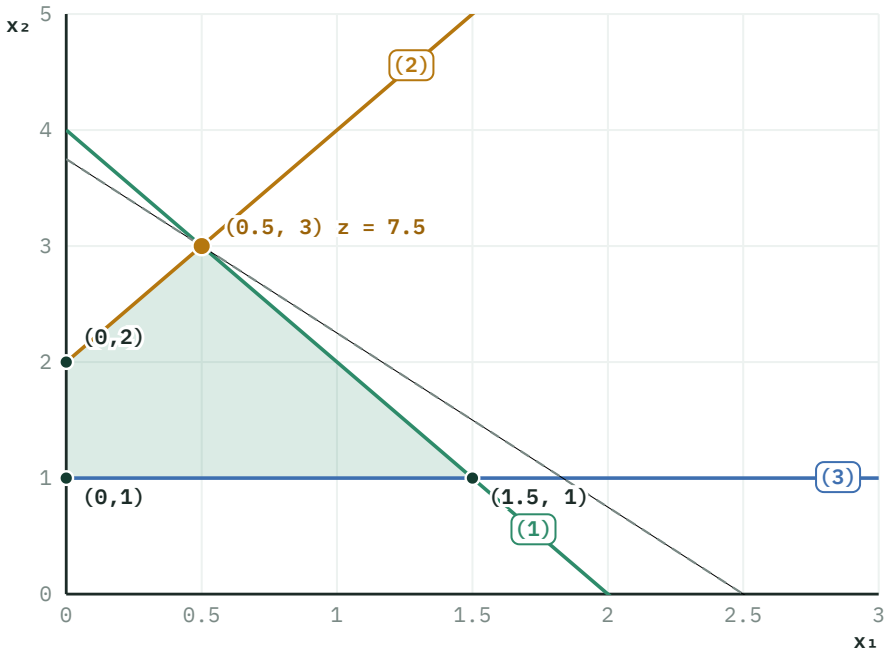
2. In the objective function row, the coefficients of the basic variables are zero.

02 Solution Q2: graphical

1 Prepare the constraints

(1)	$2x_1 + x_2 = 4$	points (0, 4) and (2, 0)
(2)	$-2x_1 + x_2 = 2$	points (0, 2) and (−1, 0)
(3)	$x_2 = 1$	horizontal line

Constraints (1) and (2) hold at $(0, 0)$, so their region is on the origin's side.
Constraint (3) is $\geq$, so its region is above the line $x_2 = 1$.



2 The corner points

Corner	Where it comes from	$z = 3x_1 + 2x_2$
(0, 1)	(3) with the x_2 axis	2
(1.5, 1)	(1) with (3)	6.5
(0.5, 3)	(1) with (2)	**7.5**
(0, 2)	(2) with the x_2 axis	4

(1) + (2): $2x_2 = 6 \rightarrow x_2 = 3$ $\qquad$ $2x_1 + 3 = 4 \rightarrow x_1 = 0.5$

$Z = 7.5$ at $x_1 = 0.5$ and $x_2 = 3$

03 Solution Q3: standard form

1 The methods — 2 M

Graphical method and Algebraic method.

2 Standard form — 6 M

$4x_2 = -7 \quad \times (-1) \quad \rightarrow \quad -4x_2 = 7$

$x_2 = x_2^+ - x_2^-$ in every place x_2 appears

$$\text{Maximum } Z = 8x_1 - 2x_2^+ + 2x_2^- + 0s_1 + 0s_2$$

$$\text{S.to} \quad 2x_1 + x_2^+ - x_2^- + s_1 = 12$$

$$x_1 - 5x_2^+ + 5x_2^- - s_2 = 6$$

$$-4x_2^+ + 4x_2^- = 7$$

$$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$$

Marking: two marks for the objective function and one for each of the four lines.

04 Solution Q4: simplex for a min problem

a Standard form 2 M

Minimize $Z = x_1 - 2x_2 + 0s_1 + 0s_2$	
S.to	$2x_1 + 3x_3 + s_1 = 1$
	$3x_1 + 2x_2 - x_3 + s_2 = 5$
	$x_1, x_2, x_3, s_1, s_2 \geq 0$

The key writes the last line as $x_1, x_2, s_1, s_2, s_3 \geq 0$, which is a typo: the problem has x_3 and no s_3.

b Simplex `8 M`

The `Z` row comes from `z − x1 + 2x2 = 0`. The problem is min, so the entering variable is the most positive entry.

Basic	x_1	x_2	x_3	s_1	s_2	Sol.	Ratio
s_1	2	0	3	1	0	1	–
s_2	3	**2**	−1	0	1	5	$\frac{5}{2}$
Z	−1	2	0	0	0	0	
s_1	2	0	**3**	1	0	1	$\frac{1}{3}$
x_2	$\frac{3}{2}$	1	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{5}{2}$	–
Z	−4	0	1	0	−1	−5	
x_3	$\frac{2}{3}$	0	1	$\frac{1}{3}$	0	$\frac{1}{3}$	
x_2	$\frac{11}{6}$	1	0	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{8}{3}$	
Z	$-\frac{14}{3}$	0	0	$-\frac{1}{3}$	−1	$-\frac{16}{3}$	

Iteration 1: the most positive entry is `2` under `x2`. The `s1` row has a zero coefficient, so it has no ratio, and the `s2` row has ratio $\frac{5}{2}$. So `s2` leaves and the pivot is `2`.

Iteration 2: a positive `1` remains under `x3`. The `s1` row has ratio $\frac{1}{3}$, and the `x2` row has a negative coefficient. So `s1` leaves and the pivot is `3`.

Optimum $z = -\frac{16}{3} = -5.33$ at $x_1 = 0$, $x_2 = \frac{8}{3} = 2.67$, $x_3 = \frac{1}{3} = 0.33$

Check: $z = 0 - 2\left(\frac{8}{3}\right) = -\frac{16}{3}$

The same tableaus in decimals (as in the answer key)

Basic	x_1	x_2	x_3	s_1	s_2	Sol.
s_1	2	0	3	1	0	1
s_2	3	**2**	−1	0	1	5
Z	−1	2	0	0	0	0
s_1	2	0	**3**	1	0	1
x_2	1.5	1	−0.5	0	0.5	2.5
Z	−4	0	1	0	−1	−5
x_3	0.67	0	1	0.33	0	0.33
x_2	1.83	1	0	0.17	0.5	2.67
Z	−4.67	0	0	−0.33	−1	−5.33

First table 2m (every row 0.5m + 0.5m for pivot element).

Second and third table 3m (1m for every row). Every mistake in calculation −0.5m.